

SIMATS ENGINEERING
ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I S. Rohith (192521127) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud.
The system must handle millions of concurrent users with minimal buffering.
Design a cloud architecture using scalability and caching techniques.
Explain how load balancing and auto-scaling can ensure smooth streaming.
Discuss cost optimization strategies for such high-demand workloads.
2. A fintech startup is deploying applications using containers across multiple cloud providers.
They face challenges in maintaining consistency and managing deployments.
Propose a solution using container orchestration and multi-cloud strategies.
Explain how service discovery and scaling can be handled efficiently.
Highlight the benefits and risks of a multi-cloud deployment model.
3. An e-learning platform experiences data loss due to accidental deletion in the cloud.
The organization needs a robust backup and disaster recovery strategy.
Design a solution using cloud-native backup, replication, and versioning.

Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.
 Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data.
 Some applications must remain on-premises due to compliance requirements.
 Design a hybrid architecture that ensures secure communication between environments.
 Explain how identity management and encryption should be implemented.
 Discuss challenges in managing hybrid cloud environments.
5. A SaaS provider needs to ensure high availability and fault tolerance for its application.
 The system must continue operating even if one region goes down.
 Propose a multi-region deployment strategy in the cloud.
 Explain how data replication and failover mechanisms should be configured.
 Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

Scenario Based

1. cloud architecture using scalability and caching
- * Deploy the video streaming platform on multiple cloud servers across different regions.
 - * Store videos in cloud object storage.
 - * Use a content delivery network (CDN) to cache video closer to users reducing buffering and latency.
 - * Use distributed databases and cloud networking for reliable data access.

Load Balancing and Auto scaling

- * Load Balancing: Distributes incoming user requests evenly across multiple servers preventing server overload.
- * Auto scaling: Automatically adds new servers when user traffic increases and removes extra servers when traffic decreases.

Cost optimization strategies.

- * Use a pay as you go pricing model to pay only for resources used.
- * Enable auto scaling to avoid paying for idle servers during low traffic.
- * Store frequently accessed videos in CDN cache and archive rarely viewed video in low cost cloud storage.

2. proposed solution -

- * use docker container to package applications with all required dependencies.
- * use Kubernetes as the container orchestration platform to deploy and manage containers across multiple cloud providers (AWS, Azure)

Service Discovery and scaling.

- * Kubernetes providing services discovery using built in DNS, allowing containers to communicate automatically without manual configuration.
- * Use Horizontal Pod Autoscaler (HPA) to automatically increase or decrease the number of container instances based on CPU or memory usage.

Benefits of multi cloud deployment

- * High availability and reliability if one cloud provider fails.
- * Avoids Vendor lock in by using multiple cloud providers
- * Better disaster recovery and business continuity.

Risk of multi cloud deployment.

- * Increased management and operational complexity
- * Higher security and compliance challenges.
- * Data synchronization issue between cloud providers.

cloud Native backup, replication, versioning solution

- * Enable automatic cloud backups to regularly backup all application data and database.
- * Use data replication to store copies of data in multiple cloud regions, ensuring availability even if one region fails.
- * Enable object versioning so deleted or modified files can be restored to previous versions.

Archiving RTO and RPO.

- * Recovery time Objective (RTO): Achieved by using replicated data automated failover and cloud backups allowing services to recover quickly.
- * Recovery point Objective (RPO): Achieved through frequent backups and continuous data replication, minimizing the amount of data lost during a disaster.

Steps to ensure business continuity.

1. Monitor backup and replication status continuously
2. Test disaster recovery plans regularly to ensure they work efficiently.
3. Store backup copies in multiple geographic regions.
4. Enable automatic failover to a secondary region during failure.
5. Periodically verify backup integrity and recovery procedure.

4. Hybrid cloud architecture

- * Store sensitive data and critical applications on the on premises data center to meet compliance requirements.
- * Deploy non sensitive applications and services in the public cloud for better scalability and cost efficiency.
- * Establish secure communication between on premises and cloud environment using a VPN or a dedicated private connection.

Identity management and encryption.

- * Implement a centralized identity and access management (IAM) system for user authentication and authorization.
- * Enable multi factor authentication (MFA) to strengthen user security.
- * Encrypt data at rest using strong encryption.

Challenges in managing hybrid cloud environment

1. Increased complexity in managing both cloud and on premises infrastructure.
2. Ensuring compliance with government security and regulatory requirement.
3. Maintaining consistent identity access control and security policies across environment.
4. Network latency and connectivity issue between cloud and on premises system.

Multi region deployment strategy.

- * Deploy the application in two or more cloud region
- * Use global load balancing to route users to the nearest healthy region.
- * Keep application servers running in all regions to ensure uninterrupted service if one region fails.

Data replication and failover.

- * Configure database replication between primary and ~~nearest healthy~~ secondary regions.
- * Enable automatic failover so traffic is redirected to the secondary region if the primary region becomes unavailable.
- * Keep backups and replicated data synchronized across all regions to prevent data loss.

Monitoring and alerting strategies.

1. Continuously monitoring CPU, memory, storage, network and application health.
2. Perform regular health checks on servers, databases and services.
3. Configure real time alerts for server failure, high resource usage and network issues.
4. Use centralized logging and monitoring dashboards to track system performance.